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Method for operating membrane filtration unit and membrane filtration unit

Abstract

The present invention relates to a method of operating a membrane filtration unit including plural hollow fiber membrane modules connected to each other in parallel, the method including: a filtration step; a collection step; and a recovery step, in which a relation of $n_{sub.1} \geq n_{sub.2} > n_{sub.3}$ is satisfied, where $n_{sub.1}$ is the number of the hollow fiber membrane modules simultaneously used in executing the filtration step, $n_{sub.2}$ is the number of the hollow fiber membrane modules simultaneously used in executing the collection step, and $n_{sub.3}$ is the number of the hollow fiber membrane modules simultaneously used in executing the recovery step.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This is the U.S. National Phase application of PCT/JP2020/007878, filed Feb. 26, 2020, which claims priority to Japanese Patent Application No. 2019-032311, filed Feb. 26, 2019, the disclosures of each of these applications being incorporated herein by reference in their entireties for all purposes.

FIELD OF THE INVENTION

(2) The present invention relates to a method of operating a membrane filtration unit and a membrane filtration unit.

BACKGROUND OF THE INVENTION

(3) Membrane filtration using separation membranes is used in various fields such as a water treatment field such as drinking water manufacture, water purification treatment, and effluent treatment, a fermentation field that involves cultivation of microbes or cultivation cells, and a food industry field. Among those kinds of membrane filtration, membrane filtration using a hollow fiber membrane module is used in many fields because of its advantages such as a large treatment liquid amount and ease of cleaning.

(4) In the food industry field, in many cases the raw liquid turbidity is higher than in the water treatment field and hence separation membranes are clogged fast in a dead-end filtration operation which is employed frequently in the water treatment field. Thus, in uses in this field, a crossflow filtration operation is performed in which the clogging of separation membranes proceeds more slowly. The crossflow filtration operation is a method in which parallel flows of raw liquid always act on the surfaces of separation membranes and part of the raw liquid is filtered. This method can greatly lower the occurrence of clogging of the separation membranes because an operation is performed while accumulation of suspended solids on the surfaces of the separation membranes is prevented by the action of flows that are parallel with the surfaces of the separation membranes.

(5) On the other hand, in the crossflow filtration operation, a large amount of raw liquid remains in the hollow fiber membrane module or piping of a membrane filtration unit because more additional pieces of equipment such as a concentrate pipe are necessary than in the dead-end filtration operation. In the food industry field, since it is important to minimize the amount of residual raw

liquid from the viewpoint of increasing the recovery rate, a membrane filtration unit for minimizing the residual raw liquid and an operation method thereof are desired.

(6) In this connection, Patent document 1 discloses a method in which pressurized gas is introduced to the raw liquid side in a tank that includes a filtration element and all of raw liquid remaining in the tank is thereby recovered to the filtrate side. Also in a filtration unit using separation membranes, a recovery method can be performed in which liquid remaining on the raw liquid side is pushed out to the filtrate side by introducing pressurized gas to the raw liquid side and liquid remaining in filtrate side piping is pushed out by passage of air through the separation membranes.

(7) Patent document 2 discloses, as another recovery method, a method in which in a membrane filtration unit including plural hollow fiber membrane modules raw liquid remaining in the hollow fiber membrane modules and piping is collected into one hollow fiber membrane module and subjected to membrane filtration there.

PATENT LITERATURE

(8) Patent document 1: JP-A-H08-066608 Patent document 2: WO 2000/018497

SUMMARY OF THE INVENTION

(9) However, the method disclosed in Patent document 1 has a problem that where separation membranes using a highly hydrophobic polymer is employed, if a method of pushing raw liquid by passing pressurized gas from the raw liquid side of the membranes to their filtrate side, the membranes are dried to cause a problem that when the liquid is filtrated again the pure water permeability becomes lower than at the beginning and hence the amount of filtered liquid obtained is reduced. On the other hand, where pressurized gas is introduced to only the raw liquid side so as not to dry the membranes, a problem arises that liquid remaining on the filtrate side of the separation membranes cannot be recovered.

(10) Also the method disclosed in Patent document 2 cannot reduce the amount of liquid remaining on the filtrate side of the separation membranes.

(11) In view of the above, an object of the present invention is to provide an operating method of a membrane filtration unit and a membrane filtration unit for reducing the amount of liquid remaining in the filtrate sides in a membrane filtration unit even in a case of using a highly hydrophobic resin as separation membranes.

(12) To attain the above object, an exemplary embodiment of the present invention provides the following methods of operating a membrane filtration unit. (1) A method of operating a membrane filtration unit, the membrane filtration unit including plural hollow fiber membrane modules connected to each other in parallel, each hollow fiber membrane module including a container which has a raw liquid inlet, a filtrate outlet, and a raw liquid outlet and is filled with hollow fiber membranes, the method including: a filtration step of introducing a raw liquid into the container from the raw liquid inlet, filtering the raw liquid through the hollow fiber membranes from a primary side to a secondary side, and leading out resulting filtrate to outside the container from the filtrate outlet; a collection step of causing the filtrate existing in the container to flow in a reverse direction through the hollow fiber membranes from the secondary side to the primary side and leading out resulting backflow liquid to outside the container from at least one of the raw liquid inlet and the raw liquid outlet; and a recovery step of introducing the backflow liquid into the container again from at least one of the raw liquid inlet and the raw liquid outlet, filtering the backflow liquid through the hollow fiber membranes from the primary side to the secondary side, and leading out resulting recovery liquid to outside the container from the filtrate outlet, in which a relation of $n_{\text{sub}.1} \geq n_{\text{sub}.2} > n_{\text{sub}.3}$ is satisfied, where $n_{\text{sub}.1}$ is the number of the hollow fiber membrane modules simultaneously used in executing the filtration step, $n_{\text{sub}.2}$ is the number of the hollow fiber membrane modules simultaneously used in executing the collection step, and $n_{\text{sub}.3}$ is the number of the hollow fiber membrane modules simultaneously used in executing the recovery step. (2) The method of operating a membrane filtration unit according to item (1), in which the hollow fiber membrane module used in executing the recovery step is different from the

hollow fiber membrane module used in executing the collection step. (3) The method of operating a membrane filtration unit according to item (2), in which the collection step and the recovery step are executed simultaneously. (4) The method of operating a membrane filtration unit according to item (3), in which the collection step and the recovery step are executed using pressurized gas and a pressure P2 of the pressurized gas used in executing the collection step and a pressure P3 of the pressurized gas used in executing the recovery step satisfy a relation of $P3 < P2$. (5) The method of operating a membrane filtration unit according to any one of items (1) to (4), in which the raw liquid inlet of the hollow fiber membrane module used in executing the recovery step is located below the raw liquid inlet of the hollow fiber membrane module that is not used in executing the recovery step in a vertical direction. (6) The method of operating a membrane filtration unit according to any one of items (1) to (5), in which the hollow fiber membranes are membranes made of a hydrophobic resin. (7) The method of operating a membrane filtration unit according to item (6), in which the hydrophobic resin includes polyvinylidene difluoride.

(13) To attain the above object, the present invention according to exemplary embodiments provides the following membrane filtration units: (8) A membrane filtration unit including plural hollow fiber membrane modules connected to each other in parallel, each hollow fiber membrane module including a container which has a raw liquid inlet, a filtrate outlet, and a raw liquid outlet and is filled with hollow fiber membranes, in which a first pressurized gas introduction pipe for introducing a pressurized gas is connected to a pipe or a tank that is connected to the raw liquid inlet or raw liquid outlet, a second pressurized gas introduction pipe for introducing a pressurized gas is connected to a pipe that is connected to the filtrate outlet, and part of the hollow fiber membrane modules includes a valve in the pipe that connects the filtrate outlet and the second pressurized gas introduction pipe. (9) The membrane filtration unit according to item (8), in which part of the hollow fiber membrane modules further includes a bypass pipe that connects the filtrate outlet and a filtrate recovery pipe or a filtrate tank, and the bypass pipe is not connected to the second pressurized gas introduction pipe. (10) The membrane filtration unit according to item (9), in which a pipe diameter of the bypass pipe is smaller than a pipe diameter of the filtrate recovery pipe.

(14) According to the present invention, the amount of liquid remaining in the filtrate sides in a membrane filtration unit can be reduced to a large extent even in a case of using a highly hydrophobic resin as separation membranes, whereby the raw liquid recovery rate can be increased. Furthermore, the time taken to recover liquid remaining in the filtrate sides of the membrane filtration unit can be shortened, whereby the operating rate of the machine can be increased remarkably.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram showing one form of a hollow fiber membrane module of a membrane filtration unit to which an operating method according to embodiments of the present invention is applied.

(2) FIG. 2 is a rough flow diagram showing one form of a membrane filtration unit to which the operating method according to embodiments of the present invention is applied.

(3) FIG. 3 is a rough flow diagram showing another form of a membrane filtration unit to which the operating method according to embodiments of the present invention is applied.

(4) FIG. 4 is a rough flow diagram showing another form of a membrane filtration unit to which the operating method according to embodiments of the present invention is applied.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

(5) Although embodiments of the present invention will be hereinafter described in detail with

reference to the drawings, the present invention is not restricted by them at all.

(6) A membrane filtration unit to which an operating method of the embodiment is applied is required to include plural hollow fiber membrane modules in each of which hollow fiber membranes fill a container having a raw liquid inlet, a filtrate outlet, and a raw liquid outlet.

(7) FIG. 1 is a schematic diagram showing one form of a hollow fiber membrane module of the membrane filtration unit to which the operating method of the embodiment is applied. FIG. 2 is a rough flow diagram showing one form of membrane filtration unit to which the operating method of the embodiment is applied. In each of plural hollow fiber membrane modules 31-36, as shown in FIG. 1, hollow fiber membranes 5 fill a container 1 having a raw liquid inlet 2, a filtrate outlet 3, and a raw liquid outlet 4.

(8) Each of the hollow fiber membrane modules 31-36 is separated into a raw liquid-side space 6 (hereinafter referred to as a primary side) and a filtrate-side space 7 (hereinafter referred to as a secondary side) by the hollow fiber membranes 5 that fill the container 1. The raw liquid inlet 2 is provided in the primary side and the filtrate outlet 3 is provided in the secondary side. Each of the hollow fiber membrane modules 31-36 has the raw liquid outlet 4 which is provided in the primary side and leads out raw liquid introduced into the container 1.

(9) In the membrane filtration unit to which the operating method of the embodiment is applied, it is required that the plural hollow fiber membrane modules (mentioned above) be connected to each other in parallel. The phrase “connected to each other in parallel” means a connection form where liquid brought by one pipe is divided into plural liquids and then introduced, as liquids having the same quality, into the plural respective hollow fiber membrane modules that are arranged adjacent to each other or liquids that are led out of the plural hollow fiber membrane modules arranged adjacent to each other are gathered into one pipe, and the plural hollow fiber membrane modules are connected together by pipes.

(10) The operating method of a membrane filtration unit according to the embodiment includes three steps described below. (1) A filtration step is a step of introducing raw liquid into the container from the raw liquid inlet, filtering the raw liquid through the hollow fiber membranes from the primary side to the secondary side, and leading out resulting filtrate to outside the container from the filtrate outlet.

(11) In the hollow fiber membrane module exemplified in FIG. 1, the raw liquid is introduced into the container 1 from the raw liquid inlet 2 and filtered through the hollow fiber membranes 5 from the primary side to the secondary side, and resulting filtrate is led out to outside the container 1 from the filtrate outlet 3. It is preferable that the filtration through the hollow fiber membranes from the primary side to the secondary side be pressure filtration. (2) A collection step is a step of causing the filtrate existing in the container to flow in the reverse direction through the hollow fiber membranes from the secondary side to the primary side and leading out resulting backflow liquid to outside the container from at least one of the raw liquid inlet and the raw liquid outlet.

(12) In the hollow fiber membrane module exemplified in FIG. 1, filtrate existing in the secondary side in the container 1 is caused to flow reversely through the hollow fiber membranes 5 from the secondary side to the primary side and resulting backflow liquid is led out to outside the container 1 from the raw liquid inlet 2 and/or the raw liquid outlet 4. (3) A recovery step is a step of introducing the backflow liquid into the container again from at least one of the raw liquid inlet and the raw liquid outlet, filtering the backflow liquid through the hollow fiber membranes from the primary side to the secondary side, and leading out resulting recovery liquid to outside the container from the filtrate outlet.

(13) In the hollow fiber membrane module exemplified in FIG. 1, the backflow liquid is introduced into the container 1 from the raw liquid inlet 2 and/or the raw liquid outlet 4 and filtered through the hollow fiber membranes 5 from the primary side to the secondary side, and resulting recovery liquid is led out to outside the container 1 from the filtrate outlet. It is preferable that the filtration through the hollow fiber membranes from the primary side to the secondary side be pressure

filtration.

(14) In the operating method of a membrane filtration unit according to the embodiment, it is necessary that a relationship of the following Inequality (I) needs to be satisfied, where $n_{sub.1}$ is the number of the hollow fiber membrane modules simultaneously used in executing the filtration step, $n_{sub.2}$ is the number of the hollow fiber membrane modules simultaneously used in executing the collection step, and $n_{sub.3}$ is the number of the hollow fiber membrane modules simultaneously used in executing the recovery step:

$$n_{sub.1} \geq n_{sub.2} > n_{sub.3} \quad (1).$$

(Membrane Filtration Unit)

(15) In the membrane filtration unit according to the embodiment, the plural hollow fiber membrane modules are connected to each other in parallel. In each of the hollow fiber membrane modules, the hollow fiber membranes fill the container having the raw liquid inlet, the filtrate outlet, and the raw liquid outlet.

(16) A first pressurized gas introduction pipe for introducing pressurized gas is connected to a pipe or a tank that is connected to the raw liquid inlet or the raw liquid outlet of the hollow fiber membrane module. A second pressurized gas introduction pipe for introducing pressurized gas is connected to a pipe that is connected to the filtrate outlet. In part of the hollow fiber membranes modules, a valve is provided on a pipe that connects the filtrate outlet and the second pressurized gas introduction pipe.

(17) It is preferable that part of the hollow fiber membrane modules further includes a bypass pipe that connects the filtrate outlet and a filtrate recovery pipe or a filtrate tank, and it is preferable that the bypass pipe is not connected to the second pressurized gas introduction pipe.

(18) In the one form of the membrane filtration unit exemplified in FIG. 2, a raw liquid tank **11** is connected to the raw liquid inlets of the hollow fiber membrane modules **31-36** by a supply liquid pipe **21**. The supply liquid pipe **21** branches in the rear of a supply pump **13** and a supply liquid valve **41** which are disposed at halfway positions of the supply liquid pipe **21** and are connected to the hollow fiber membrane modules **31-36** in parallel. The filtrate outlets of the hollow fiber membrane modules **31-36** are connected to a filtrate tank **12** in parallel by a filtrate pipe **22**. The raw liquid outlets of the hollow fiber membrane modules **31-36** are connected to the raw liquid tank **11** in parallel by a concentrate pipe **23**. A filtrate valve **42** is disposed at a halfway position of the filtrate pipe **22** and a concentrate valve **43** is disposed at a halfway position of the concentrate pipe **23**.

(19) Furthermore, a filtrate division valve **44** for separating filtrate led out of the hollow fiber membrane modules **31-35** from filtrate led out of the hollow fiber membrane module **36** so that they do not come into contact with each other is disposed at a halfway position of the filtrate pipe **22**.

(20) A second pressurized gas introduction pipe **26** for introducing gas at a halfway position of which a filtrate gas introduction valve **49** is disposed is connected to the filtrate pipe **22** before the filtrate valve **42**. Likewise, a first pressurized gas introduction pipe **25** for introducing gas at a halfway position of which a concentrate gas introduction valve **48** is disposed is connected to the concentrate pipe **23** before the concentrate valve **43**.

(21) The filtrate tank **12** is further connected to the filtrate pipe **22** by a backwash pipe **27**. A backwash pump **14** and a backwash valve **46** for sending filtrate to the hollow fiber membrane modules **31-36** from the filtrate tank **12** are disposed at a halfway position of the backwash pipe **27**.

(22) A filtrate bypass pipe **24** is connected between the filtrate outlet of the hollow fiber membrane module **36** of the filtrate pipe **22** and the filtrate division valve **44**. The other end of the filtrate bypass pipe **24** is connected to a filtrate recovery pipe **28** provided between the filtrate valve **42** of the filtrate pipe **22** and the filtrate tank **12**. A filtrate bypass valve **45** is disposed at a halfway position of the filtrate bypass pipe **24**. To reduce the amount of residual liquid, it is preferable that the filtrate bypass pipe **24** be thinner than the filtrate pipe **22**.

(23) A backwash bypass valve **47** for separation is disposed at a halfway position of the filtrate pipe **22** to prevent filtrate sent from the filtrate tank **12** by the backwash pump **14** from being introduced into the hollow fiber membrane modules **31-35**.

(24) The operating method of the membrane filtration unit according to the embodiment will be described using specific examples while referring to FIG. 2 mainly.

(25) (Operating Method of Membrane Filtration Unit: Filtration Step)

(26) In a filtration step, for example, raw liquid is supplied to the hollow fiber membrane modules **31-36** by the supply pump **13** and a crossflow filtration operation is performed. In the crossflow filtration operation, raw liquid is introduced into the primary side (raw liquid-side space **6**) in the container **1** from the raw liquid inlets **2** of the hollow fiber membrane modules **31-36**, moves through the primary side in the container **1** as flows that are parallel with the surfaces of the hollow fiber membranes **5**, and part of the raw liquid is filtered into the secondary side (filtrate-side space **7**) in the container **1** preferably under pressurization. Filtrate is led out of the container **1** from the filtrate outlets **3** and sent to the filtrate tank **12** via the filtrate pipe **22**. On the other hand, unfiltered raw liquid is led out from the raw liquid outlets **4** as concentrate and sent to the raw liquid tank **11** via the concentrate pipe **23**. At this time, the supply liquid valve **41** is open, the filtrate valve **42** is open, the concentrate valve **43** is open, the filtrate division valve **44** is open, the filtrate bypass valve **45** is closed, the backwash valve **46** is closed, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48** is closed, and the filtrate gas introduction valve **49** is closed.

(27) In many cases, the crossflow filtration operation is performed until raw liquid has been filtered for a prescribed time or by a prescribed amount and the amount of liquid remaining in the raw liquid tank **11** becomes small so that operation of the supply pump **13** become difficult. Since in this case residual liquid exists in the supply liquid pipe **21**, the primary side and the secondary side in the containers **1** of the hollow fiber membrane modules **31-36**, the filtrate pipe **22**, and the concentrate pipe **23**, the residual liquid is required to be recovered into the filtrate tank **12** as much as possible to increase the recovery rate.

(28) To this end, it is preferable that after operation of the supply pump **13** become difficult, residual liquid existing in the primary sides in the containers **1** of the hollow fiber membrane modules and the concentrate pipe **23** located between the raw liquid outlets **4** of the hollow fiber membrane modules **31-36** and the concentrate valve **43** be subjected to gas pressure filtration by introducing pressurized gas from the first pressurized gas introduction pipe **25**. At this time, the supply liquid valve **41** is closed, the filtrate valve **42** is open, the concentrate valve **43** is closed, the filtrate division valve **44** is open, the filtrate bypass valve **45** is closed, the backwash valve **46** is closed, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48** is open, and the filtrate gas introduction valve **49** is closed.

(29) As a result of the gas pressure filtration, the concentrate pipe **23** and the primary sides in the containers **1** of the hollow fiber membrane modules **31-36** become empty. Filtrate is led out of the containers **1** from the filtrate outlets **3** and sent to the filtrate tank **12** via the filtrate pipe **22**. On the other hand, residual liquid exists in the supply liquid pipe **21**, the secondary sides in the containers **1** of the hollow fiber membrane modules **31-36**, and the filtrate pipe **22**.

(30) In the filtration process, a dead-end filtration operation may be performed instead of the crossflow filtration operation. In the dead-end filtration operation, although raw liquid is introduced into the primary sides in the containers **1** from the filtrate outlets **3** of the hollow fiber membrane modules, the raw liquid that has been introduced into the primary sides in the containers **1** is not led out from the raw liquid outlets **4** and is all filtered from the primary sides to the secondary sides under pressurization. Filtrate is led out of the containers **1** from the filtrate outlets **3** and sent to the filtrate tank **12** via the filtrate pipe **22** as in the case of the crossflow filtration operation. At this time, the supply liquid valve **41** is open, the filtrate valve **42** is open, the concentrate valve **43** is closed, the filtrate division valve **44** is open, the filtrate bypass valve **45** is closed, the backwash valve **46** is closed, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48**

is closed, and the filtrate gas introduction valve **49** is closed.

(31) In the one form of the membrane filtration unit exemplified in FIG. 2, concentrate is sent to the raw liquid tank **11** in many cases. Alternatively, all or part of concentrate is sent to a downstream stage of the supply pump **13**. This configuration can lower the motive power consumption of the supply pump **13**. In this case, it is preferable that a circulation pump be provided downstream of the supply pump **13** so that concentrate is sent between the supply pump **13** and the circulation pump.

(32) Whereas a common method for causing pressurization in the membrane filtration unit is to use the supply pump **13** as shown in FIG. 2, gas pressure filtration may be performed by introducing pressurized gas into the raw liquid tank **11**, the supply liquid pipe **21**, or the concentrate pipe **23**. In the food industry field, it is preferable to use, for example, nitrogen gas or carbon dioxide gas as pressurized gas and it is even preferable to use sterilized gas.

(33) In the filtration step, the above-described crossflow filtration operation or the dead-end filtration operation may be performed singly; alternatively, the crossflow filtration operation and the dead-end filtration operation may be performed in combination. It is also possible to perform filtration by causing suction in the secondary sides in the container **1** instead of causing pressurization in the primary sides in the container **1**. In this case, for example, a suction pump may be provided at a halfway position of the filtrate pipe **22**.

(34) If the filtration step is continued for a long time, the clogging of the hollow fiber membranes **5** proceeds and thus the membrane filtration amount may decrease or the pressure necessary for the filtration may increase. To avoid such a situation, it is preferable to clean the hollow fiber membrane modules **31-36** on a regular basis.

(35) A preferable method for cleaning the hollow fiber membrane modules **31-36** is backwashing of washing away suspended solids accumulated inside or on the surfaces of the hollow fiber membranes **5** by reverse filtration of sending filtrate to the hollow fiber membrane modules **31-36** by the backwash pump **14** and then pressuring filtrate from the secondary side to the primary side in each container **1**. At this time, the supply liquid valve **41** is closed, the filtrate valve **42** is closed, the concentrate valve **43** is open, the filtrate division valve **44** is open, the filtrate bypass valve **45** is closed, the backwash valve **46** is open, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48** is closed, and the filtrate gas introduction valve **49** is closed. Alternatively, backwashing may be performed while raw liquid is supplied to the hollow fiber membrane modules **31-36** by opening the supply liquid valve **41** and causing the supply pump **13** to operate.

(36) Although there are no limitations on the number n.sub.1 of hollow fiber membrane modules that are used simultaneously to execute the filtration step, it is preferable to use all of the plural hollow fiber membrane modules that are connected to each other in parallel.

(37) (Membrane Filtration Unit Operating Method: Collection Step)

(38) In the collection step, first, manipulations are performed to cause filtrate existing in the secondary sides in the containers **1** of the hollow fiber membrane modules **31-36**, that is, residual liquid, to flow in the reverse direction to the primary sides of the hollow fiber membranes and thereby obtain backflow liquid. More specifically, pressurized gas is introduced from the second pressurized gas introduction pipe **26** to pressurize the secondary sides in the containers **1** of the hollow fiber membrane modules **31-36**, whereby filtrate existing in the secondary sides in the container **1** and the filtrate pipe **22** located between the filtrate outlets **3** of the hollow fiber membrane modules **31-36** and the filtrate valve **42** and the backwash valve **46** is caused to flow in the reverse direction to the primary sides. At this time, the supply liquid valve **41** is closed, the filtrate valve **42** is closed, the concentrate valve **43** is open, the filtrate division valve **44** is open, the filtrate bypass valve **45** is closed, the backwash valve **46** is closed, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48** is closed, and the filtrate gas introduction valve **49** is open.

(39) Backflow liquid is accumulated in the primary sides in the containers **1** or led out from the raw liquid inlet **2** and accumulated in the pipe to which the raw liquid inlets **2** are connected. Depending on the capacities of the secondary side of each container **1** and the filtrate pipe **22**, there may occur an event that not all backflow liquid is accommodated in the primary sides in the containers **1** and part of the backflow liquid overflows from the raw liquid outlets **4** to flow into the concentrate pipe **23**. Backflow liquid may be led out from the raw liquid inlets **2** and the raw liquid outlet **44** at the same time.

(40) Next, manipulations are performed to lead out backflow liquid existing in the primary sides in the containers **1** of the hollow fiber membrane modules **31-35** to outside the containers **1** from the raw liquid inlets **2** by introducing pressurized gas from the first pressurized gas introduction pipe **25**. At this time, the supply liquid valve **41** is closed, the filtrate valve **42** is closed, the concentrate valve **43** is closed, the filtrate division valve **44** is closed, the filtrate bypass valve **45** is open, the backwash valve **46** is closed, the backwash bypass valve **47** is closed, the concentrate gas introduction valve **48** is open, and the filtrate gas introduction valve **49** is closed.

(41) Backflow liquid may be led out of the containers **1** from the raw liquid outlets **4** simultaneously with or instead of that backflow liquid is led out of the containers **1** from the raw liquid inlets **2**. Where backflow liquid is led out of the containers **1** from the raw liquid outlets **4**, pressurized gas is introduced from the pipe to which the raw liquid inlets **2** are connected. Where backflow liquid is led out of the containers **1** from both of the raw liquid inlets **2** and the raw liquid outlets **4**, pressurized gas is introduced from the second pressurized gas introduction pipe **26** which is connected to the filtrate outlets **3**.

(42) Since the backwash bypass valve **47** is closed, even if pressurized gas is introduced from the first pressurized gas introduction pipe **25**, almost no part of backflow liquid existing in the primary sides in the containers **1** of the hollow fiber membrane modules **31-35** is filtered into the secondary sides and, instead, the backflow liquid is led out to outside the containers **1** from the raw liquid inlets **2**, sent to the hollow fiber membrane module **36** via the supply liquid pipe **21**, and collected there.

(43) At the end of the above-exemplified collection step, whereas the secondary sides in the containers **1** of the hollow fiber membrane modules **31-36** and the filtrate pipe **22** located between the filtrate outlets **3** of the hollow fiber membrane modules **31-36** and the filtrate valve **42** and the backwash valve **46** are empty, residual liquid remains in the supply liquid pipe **21**, the primary sides in the containers **1** of the hollow fiber membrane modules **31-36**, and the concentrate pipe **23**.

(44) Manipulations for obtaining backflow liquid is not limited to the above-described manipulations of a method of introducing pressurized gas from the second pressurized gas introduction pipe **26**, and can be manipulations of a method of causing filtrate to flow in the reverse direction from the secondary sides to the primary sides in the containers **1** by opening the filtrate pipe **22** to the atmospheric pressure. At this time, for example, following settings can be made: the supply liquid valve **41** is closed, the filtrate valve **42** is open, the concentrate valve **43** is open, the filtrate division valve **44** is open, the filtrate bypass valve **45** is closed, the backwash valve **46** is closed, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48** is closed, and the filtrate gas introduction valve **49** is closed. Where the filtrate tank **12** is open to the atmospheric pressure, the filtrate pipe **22** is also open to the atmospheric pressure and hence filtrate in the filtrate pipe **22** can be caused to flow in the reverse direction. Alternatively, a method of opening the filtrate gas introduction valve **49** to the atmospheric pressure may be employed.

(45) The number **112** of hollow fiber membrane modules that are used simultaneously in executing the collection step may be the same as the number $n_{sub.1}$ of hollow fiber membrane modules that are used in executing the filtration step. On the other hand, where the collection step is executed also using hollow fiber membrane modules that are used in executing the downstream recovery step, since filtrate to be recovered is returned to the primary sides in the containers **1** as backflow liquid, the amount of backflow liquid to be filtrated is increased in the recovery step and hence the

recovery step takes long time. Thus, it is preferable to execute the collection step using hollow fiber membrane modules that are used in executing the filtration step but are not used in executing the recovery step. In this case, if a method of introducing pressurized gas from the second pressurized gas introduction pipe **26** is employed, following settings are made: the supply liquid valve **41** is closed, the filtrate valve **42** is closed, the concentrate valve **43** is open, the filtrate division valve **44** is closed, the filtrate bypass valve **45** is closed, the backwash valve **46** is closed, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48** is closed, and the filtrate gas introduction valve **49** is open. The number $n_{sub.2}$ of hollow fiber membrane modules that are used simultaneously in executing the collection step becomes smaller than the number $n_{sub.1}$ of hollow fiber membrane modules that are used in executing the filtration step. That is, a relationship $n_{sub.1} \geq n_{sub.2}$ is satisfied.

(46) (Membrane Filtration Unit Operating Method: Recovery Step)

(47) In the recovery step, backflow liquid is again introduced into the containers **1** from at least one of the raw liquid inlets **2** and the raw liquid outlets **4** and filtration is performed from the primary sides of the hollow fiber membranes **5** to their secondary sides. For example, if backflow liquid partially remains in the containers **1**, this backflow liquid and backflow liquid that has been led out of the containers **1** are collected into the hollow fiber membrane module **36** and pressure filtration is performed by introducing pressurized gas from the first pressurized gas introduction pipe **25**. At this time, the supply liquid valve **41** is closed, the filtrate valve **42** is closed, the concentrate valve **43** is closed, the filtrate division valve **44** is closed, the filtrate bypass valve **45** is open, the backwash valve **46** is closed, the backwash bypass valve **47** is closed, the concentrate gas introduction valve **48** is open, and the filtrate gas introduction valve **49** is closed.

(48) In the hollow fiber membrane module **36**, collected backflow liquid and raw liquid are introduced from the raw liquid inlet **2** into the primary side in the container **1** and filtered into the secondary side in the container **1** by gas pressure filtration. Recovery liquid obtained by the filtration is led out of the container from the filtrate outlet **3** and sent to the filtrate tank **12** via the filtrate bypass valve **45**. The gas pressure filtration is continued until no filtrate comes to be led out of the container **1** from the filtrate outlet **3** of the hollow fiber membrane module **36**.

(49) At the end of the above-exemplified recovery step, the concentrate pipe **23** between the first pressurized gas introduction pipe **25** and the hollow fiber membrane modules **31-36**, the primary sides in the containers **1** of the hollow fiber membrane modules **31-36**, the secondary sides in the containers **1** of the hollow fiber membrane modules **31-35**, and the filtrate pipe **22** between the filtrate outlets **3** of the hollow fiber membrane modules **31-36** and the filtrate division valve **44**, the filtrate valve **42** and the backwash valve **46** become empty and filtrate remains only in the supply liquid pipe **21**, the secondary side in the container **1** of the hollow fiber membrane module **36**, and the filtrate bypass pipe **24**.

(50) Since as described above the number m of hollow fiber membrane modules that are used simultaneously in executing the recovery step is smaller than the number $n_{sub.2}$ of hollow fiber membrane modules that are used simultaneously in executing the collection step, residual liquid existing in the secondary sides in the containers **1** and the filtrate pipe **22** can be recovered via the hollow fiber membrane modules that are used in executing the recovery step. There are no particular limitations on the number $n_{sub.3}$ of hollow fiber membrane modules that are used in executing the recovery step, it may be determined as appropriate on the basis of the amount of liquid remaining in the membrane filtration unit, the time taken by the recovery step, and other factors as long as the relationship $n_{sub.3} < n_{sub.2}$ is satisfied. However, to further reduce the amount of liquid remaining in the membrane filtration unit, it is preferable that the recovery step be performed using only one hollow fiber membrane module ($n_{sub.3}=1$).

(51) If the recovery step is continued for a long time, there may occur a phenomenon that the clogging of the hollow fiber membranes **5** proceeds and the membrane filtration rate decreases. To avoid such a situation, it is preferable to clean the hollow fiber membrane modules on a regular

basis. Example methods for cleaning only the hollow fiber membrane module **36** that is used in executing the recovery step include backwashing performed by causing the backwash pump **14** to operate with the following settings: the supply liquid valve **41** is closed, the filtrate valve **42** is closed, the concentrate valve **43** is open, the filtrate division valve **44** is open, the filtrate bypass valve **45** is closed, the backwash valve **46** is open, the backwash bypass valve **47** is closed, the concentrate gas introduction valve **48** is closed, and the filtrate gas introduction valve **49** is closed. (52) To push out recovery liquid remaining in the filtrate pipe **22** and the filtrate bypass pipe **24**, it is preferable to send liquid remaining in these pipes to the filtrate tank **12** by introducing pressurized gas from the second pressurized gas introduction pipe **26**. Although the amount of liquid remaining in these pipes is slight, this makes it possible to further increase the recovery rate of raw liquid. To recover liquid remaining in the filtrate bypass pipe **24**, the following settings are made: the supply liquid valve **41** is closed, the filtrate valve **42** is closed, the concentrate valve **43** is closed, the filtrate division valve **44** is open, the filtrate bypass valve **45** is open, the backwash valve **46** is closed, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48** is closed, and the filtrate gas introduction valve **49** is open. To recover liquid remaining in the filtrate pipe **22**, the following settings are made: the supply liquid valve **41** is closed, the filtrate valve **42** is open, the concentrate valve **43** is closed, the filtrate division valve **44** is closed, the filtrate bypass valve **45** is closed, the backwash valve **46** is closed, the backwash bypass valve **47** is closed, the concentrate gas introduction valve **48** is closed, and the filtrate gas introduction valve **49** is open.

(53) Where it is difficult to cause all filtrate remaining on the secondary sides of the hollow fiber membranes to flow in the reverse direction into the primary sides by executing the collection step only once, a set of the collection step and the recovery step may be executed plural times repeatedly.

(54) Furthermore, it is preferable to use different sets of hollow fiber membrane modules in executing the collection step and in executing the recovery step. In this case, it is possible to execute the collection step and the recovery step simultaneously. More specifically, the collection step and the recovery step can be executed simultaneously by, for example, introducing pressurized gas from the second pressurized gas introduction pipe **26** and the first pressurized gas introduction pipe **25** at the same time. In this case, the following settings are made: the supply liquid valve **41** is closed, the filtrate valve **42** is closed, the concentrate valve **43** is closed, the filtrate division valve **44** is closed, the filtrate bypass valve **45** is open, the backwash valve **46** is closed, the backwash bypass valve **47** is open, the concentrate gas introduction valve **48** is open, and the filtrate gas introduction valve **49** is open. To prevent backflow liquid from being filtered again into the secondary sides in the containers **1**, it is preferable to set the pressure P2 of pressurized gas that is introduced from the second pressurized gas introduction pipe **26** to execute the collection step higher than the pressure P3 of pressurized gas that is introduced from the first pressurized gas introduction pipe **25** to execute the recovery step.

(55) Since the filtrate division valve **44** is closed in the above-mentioned settings, the pressurized gas introduced from the second pressurized gas introduction pipe **26** causes filtrate existing in the secondary sides in the containers **1** of the hollow fiber membrane modules **31-35** and the filtrate pipe **22** between the filtrate outlets **3** of the hollow fiber membrane modules **31-35** and the filtrate division valve **44**, the filtrate valve **42**, and the backwash valve **46** to flow in the reverse direction to the primary sides in the containers **1** of the hollow fiber membrane modules **31-35**, whereby backflow liquid is obtained. At the same time, the pressurized gas introduced from the first pressurized gas introduction pipe **25** causes backflow liquid existing in the primary sides in the containers **1** of the hollow fiber membrane modules **31-35** to be led out from the raw liquid inlets **2**, sent to the hollow fiber membrane module **36** via the supply liquid pipe **21** and collected there. In the hollow fiber membrane module **36**, the collected backflow liquid is introduced into the primary side in the container **1** from the raw liquid inlet **2** and filtered into the secondary side in the

container **1** by gas pressure filtration. Since the filtrate division valve **44** is closed and the filtrate bypass valve **45** is open, the filtered recovery liquid is sent to the tank via the filtrate bypass pipe **24** and recovered there. At this time, setting P2 higher than P3 is preferable because filtration is performed only in the hollow fiber membrane module **36** which is a module for recovery while refiltration is not performed in the hollow fiber membrane modules **31-35**. It is even preferable that a relationship $(P2-P3)>10\text{ kPa}$ be satisfied and it is further preferable that a relationship $(P2-P3)>20\text{ kPa}$ be satisfied.

(56) Since the above-described method makes it possible to execute the collection step and the recovery step simultaneously, the time to execute the collection step and the recovery step can be shortened and the operating rate of the membrane filtration unit can be increased.

(57) The method for reducing the amount of liquid remaining in the secondary sides in the containers **1** and the filtrate pipe **22** has been described so far, the gas pressure filtration using pressurized gas cannot filter raw liquid located below the lower ends of the hollow fiber membranes **5** which can come into contact with raw liquid. As a result, in the membrane filtration unit in which all the hollow fiber membrane modules **31-36** which are connected to each other in parallel as shown in FIG. **2** are located above the supply liquid pipe **21** in the vertical direction, raw liquid existing below the lower ends of the hollow fiber membranes **5** in the primary sides in the containers **1** of the hollow fiber membrane modules **31-36** and raw liquid existing in the supply liquid pipe **21** are not filtered and become residual liquid.

(58) In the operating method of the membrane filtration unit according to the embodiment, the amount of liquid remaining in the supply liquid pipe **21** can be reduced by disposing the raw liquid inlet **2** of the hollow fiber membrane module **36** to be used in the recovery step below the raw liquid inlets **2** of the hollow fiber membrane modules **31-35** not to be used in the recovery step in the vertical direction as shown in FIG. **3**.

(59) In a recovery step using a membrane filtration unit shown in FIG. **3**, for example, backflow liquid and raw liquid are collected into the hollow fiber membrane module **36** by introducing pressurized gas from the first pressurized gas introduction pipe **25** and the backflow liquid and the raw liquid are subjected to gas pressure filtration in the hollow fiber membrane module **36**. In this case, since the raw liquid inlet **2** of the hollow fiber membrane module **36** is lower than the raw liquid inlets **2** of the hollow fiber membrane modules **31-35** in the vertical direction, backflow liquid existing below the lower ends of the hollow fiber membranes **5** in the primary sides in the containers **1** of the hollow fiber membrane modules **31-35** and raw liquid existing in the supply liquid pipe **21** settle so as to have the same liquid surface level in the vertical direction. As a result, in the hollow fiber membrane module **36**, the liquid surface height in the primary side in the container **1** comes to be relatively higher than in the hollow fiber membrane modules **31-35** and hence wider portions of the hollow fiber membranes **5** in the hollow fiber membrane module **36** come to be immersed in the backflow liquid. At the same time, since pressurized gas is introduced from the first pressurized gas introduction pipe **25**, part of the backflow liquid in contact with the hollow fiber membranes **5** of the hollow fiber membrane module **36** is filtered to the secondary side in the container **1** and recovered.

(60) To enhance the effect described in the above example, it is preferable to dispose the raw liquid inlet **2** of the hollow fiber membrane module **36** lower than the lower end of the supply liquid pipe **21** and it is even preferable to dispose the lower ends of the hollow fiber membranes **5** of the hollow fiber membrane module **36** below the lower end of the supply liquid pipe **21**. In this case, most of liquid remaining in the supply liquid pipe **21** can be sent to the hollow fiber membrane module **36**.

(61) The operating method of the membrane filtration unit according to the embodiment can also be applied to another form of membrane filtration unit shown in FIG. **4**. In the one form of membrane filtration unit exemplified in FIG. **4**, the raw liquid tank **11** is connected to the raw liquid inlets **2** of the hollow fiber membrane modules **31-36** by the supply liquid pipe **21**. The

supply liquid pipe **21** branches downstream of the supply pump **13** which is disposed at a halfway position of the supply liquid pipe **21**, and is connected to the hollow fiber membrane modules **31-36** in parallel. The filtrate outlets **3** of the hollow fiber membrane modules **31-36** are connected to the filtrate tank **12** in parallel by the filtrate pipe **22**. The raw liquid outlets **4** of the hollow fiber membrane modules **31-36** are connected to the raw liquid tank **11** in parallel by the concentrate pipe **23**. The supply liquid valve **41** is disposed at a halfway position of the supply liquid pipe **21** and the concentrate valve **43** is disposed at a halfway position of the concentrate pipe **23**.

(62) Furthermore, a filtrate division valve **44** which separates filtrate led out of the hollow fiber membrane modules **31-35** and filtrate led out of the hollow fiber membrane module **36** from each other so that they do not come into contact with each other is disposed in the filtrate pipe **22**.

(63) The second pressurized gas introduction pipe **26** for introducing gas is connected between the filtrate division valve **44** in the filtrate pipe **22** and the filtrate outlets **3** of the hollow fiber membrane modules **31-35**, and the filtrate gas introduction valve **49** is disposed at a halfway position of the second pressurized gas introduction pipe **26**. The first pressurized gas introduction pipe **25** for introducing gas is connected between the concentrate valve **43** in the concentrate pipe **23** and the raw liquid outlets **4** of the hollow fiber membrane modules **31-35**, and the filtrate gas introduction valve **48** is disposed at a halfway position of the first pressurized gas introduction pipe **25**.

(64) The backwash pipe **27** connecting the filtrate tank **12** and the filtrate pipe **22** is provided, and the backwash pump **14** and the backwash valve **46** for sending filtrate from the filtrate tank **12** to the hollow fiber membrane modules **31-36** are disposed at halfway positions of the backwash pipe **27**.

(65) When the membrane filtration unit shown in FIG. **4** is used, since it does not have the filtrate bypass pipe **24** which is provided in the membrane filtration unit shown in FIG. **2**, the filtration unit is simpler in configuration though the amount of residual liquid existing in the filtrate pipe **22** at the end of the recovery step may increase.

(66) As another form of membrane filtration unit according to the embodiment, another hollow fiber membrane module may be connected in series downstream of the raw liquid outlet **4** of the hollow fiber membrane modules connected in parallel. The expression "connected in series" as used herein means such a connection form that at least part of raw liquid introduced into one hollow fiber membrane module is led out of this hollow fiber membrane module and introduced into another hollow fiber membrane module connected to downstream of it. In this case, for example, of two hollow fiber membrane modules connected to each other in series, the downstream hollow fiber membrane module is possibly provided with the raw liquid inlet **2** at the top in the vertical direction and with the raw liquid outlet **4** at the bottom in the vertical direction. In such a membrane filtration unit according to the embodiment, in the downstream one of the series-connected hollow fiber membrane modules, backflow liquid is lead out from the raw liquid outlet **4** of the hollow fiber membrane modules in the collection step and the backflow liquid is introduced into the hollow fiber membrane module from its raw liquid outlet **4** in the recovery step.

(67) It is preferable that the material of hollow fiber membranes to which the membrane filtration unit operating method according to the embodiment is applied be a hydrophobic resin. Where membranes made of a hydrophilic resin are used, even when the membranes are dried by supplying pressurized gas to the filtrate side from the raw liquid side, the membranes turn completely wet by supplying liquid again. Thus, no deterioration occurs in water permeability; originally it is not necessary to apply the operating method of a membrane filtration unit according to the embodiment.

(68) On the other hand, where membranes made of a hydrophobic resin are used, when the membranes are dried by supplying pressurized gas to the filtrate side from the raw liquid side, the membranes do not turn wet completely even if liquid is supplied again; accordingly the water permeability lowers. The embodiment is an operating method that is suitable for a membrane

filtration unit using such separation membranes that cannot be dried.

(69) Example hydrophobic resins are fluorine-based resins and olefin-based resins. Resins that are larger than or equal to 80° in contact angle are preferable and resins that are larger than or equal to 90° in contact angle are even preferable. Plural kinds of resins may be contained as long as the contact angle is in this range.

(70) Among hydrophobic resins, fluorine-based resins are preferable and polyvinylidene difluoride is even preferable from the viewpoint of strength.

(71) As for measurement of a contact angle, after a separation membrane shaped like a hollow fiber membrane is cleaned sufficiently by ethanol or pure water and vacuum-dried sufficiently, the surface of the separation membrane is made flat. An average value of contact angles measured after a lapse of 2 to 5 seconds from dropping of a pure water droplet onto the surface of separation membrane is employed as a contact angle. A contact angle is determined by the $\theta/2$ method. The $\theta/2$ method is a method of obtaining a contact angle θ from an angle ($\theta/2$) that is formed by a straight line connecting a left or right end point of a water droplet and its top and the membrane surface.

(72) To prevent drying of the hollow fiber membranes, it is preferable that the pressure for introducing the pressurized gas be lower than or equal to a bubble point pressure. The bubble point pressure is determined by, for example, introducing pressurized gas from one end of a hollow fiber membrane and increasing the pressure gradually after the hollow fiber membrane was wetted by ethanol or the like and then pores of the hollow fiber membrane were filled with prescribed liquid, and regarding a minimum pressure at which generation of bubbles from the surface at the other end is detected as the bubble point pressure. It is preferable that the liquid for making the hollow fiber membrane wet in measuring a bubble point pressure be the same liquid as raw liquid to be filtered actually or a liquid that is equivalent in surface tension to the raw liquid.

EXAMPLES

(73) (Preparation of Hollow Fiber Membrane Module)

(74) A cartridge-type hollow fiber membrane module as exemplified in FIG. 1 was prepared using 7,000 PVDF hollow fiber membranes (HFM membrane) manufactured by Toray Industries, Inc. The two ends of the hollow fiber membranes 5 were fixed by potting portions 8 formed by centrifugal potting and the hollow fiber membranes 5 were sealed at one of the two ends. The inner diameter of the container 1 was 159 mm. The contact angle of the hollow fiber membranes 5 was 95° and the bubble point pressure was 180 kPa.

Example 1

(75) A membrane filtration unit exemplified in FIG. 2 was constructed by using six hollow fiber membrane modules prepared in such a manner that their longitudinal direction coincides with the vertical direction.

(76) The filtration step was executed by a crossflow filtration operation by sending raw liquid from the raw liquid tank 11 to the hollow fiber membrane modules 31-36 by the supply pump 13. At this time, the supply liquid valve 41 was open, the filtrate valve 42 was open, the concentrate valve 43 was open, the filtrate division valve 44 was open, the filtrate bypass valve 45 was closed, the backwash valve 46 was closed, the backwash bypass valve 47 was open, the concentrate gas introduction valve 48 was closed, and the filtrate gas introduction valve 49 was closed. The crossflow flow rate and the filtration flow rate were set at 103 m.sup.3/hr and 7.7 m.sup.3/hr, respectively.

(77) Backwashing was performed in the middle of the filtration step to clean clogging of the hollow fiber membranes 5. More specifically, filtrate liquid was sent to the hollow fiber membrane modules 31-36 from the filtrate tank 12 by the backwash pump 14. At this time, the supply liquid valve 41 was closed, the filtrate valve 42 was closed, the concentrate valve 43 was open, the filtrate division valve 44 was open, the filtrate bypass valve 45 was closed, the backwash valve 46 was open, the backwash bypass valve 47 was open, the concentrate gas introduction valve 48 was

closed, and the filtrate gas introduction valve **49** was closed. The backwash flow rate was set at 10.4 m.sup.3/hr.

(78) In the filtration step, a cycle consisting of the above-mentioned crossflow filtration operation for 9 minutes and the above-mentioned backwashing for 1 minute was performed repeatedly. The crossflow filtration operation was stopped when the amount of raw liquid became small and liquid sending was completed to the lower limit level of the raw liquid tank **11**, and then gas pressure filtration was started by introducing nitrogen gas of 100 kPa from the first pressurized gas introduction pipe **25**. At this time, the supply liquid valve **41** was closed, the filtrate valve **42** was open, the concentrate valve **43** was closed, the filtrate division valve **44** was open, the filtrate bypass valve **45** was closed, the backwash valve **46** was closed, the backwash bypass valve **47** was open, the concentrate gas introduction valve **48** was open, and the filtrate gas introduction valve **49** was closed. The gas pressure filtration was performed until there was no filtrate led out from the filtrate outlet **3** of each hollow fiber membrane module. Subsequently, the first pressurized gas introduction pipe **25** was opened to atmospheric pressure and set the concentrate pipe **23** at normal pressure to finish the filtrate step.

(79) In the collection step, nitrogen gas of 100 kPa was introduced from the second pressurized gas introduction pipe **26**, whereby filtrate existing in the containers was caused to flow in the reverse direction from the secondary sides to the primary sides and backflow liquid was obtained. At this time, the supply liquid valve **41** was closed, the filtrate valve **42** was closed, the concentrate valve **43** was open, the filtrate division valve **44** was open, the filtrate bypass valve **45** was closed, the backwash valve **46** was closed, the backwash bypass valve **47** was open, the concentrate gas introduction valve **48** was closed, and the filtrate gas introduction valve **49** was open. The pressurization was continued for 1 minute.

(80) Subsequently, nitrogen gas of 100 kPa was introduced from the first pressurized gas introduction pipe **25**. At this time, the supply liquid valve **41** was closed, the filtrate valve **42** was closed, the concentrate valve **43** was closed, the filtrate division valve **44** was closed, the filtrate bypass valve **45** was open, the backwash valve **46** was closed, the backwash bypass valve **47** was closed, the concentrate gas introduction valve **48** was open, and the filtrate gas introduction valve **49** was closed. As a result, backflow liquid existing in the primary sides in the containers **1** of the hollow fiber membrane modules **31-35** was led out of the containers **1** from the raw liquid inlets.

(81) In the recovery step, the backflow liquid was introduced into the container **1** of the hollow fiber membrane module **36** via the supply liquid pipe **21**. Then pressure filtration was performed by introducing nitrogen gas of 100 kPa from the first pressurized gas introduction pipe **25**. At this time, the supply liquid valve **41** was closed, the filtrate valve **42** was closed, the concentrate valve **43** was closed, the filtrate division valve **44** was closed, the filtrate bypass valve **45** was open, the backwash valve **46** was closed, the backwash bypass valve **47** was closed, the concentrate gas introduction valve **48** was open, and the filtrate gas introduction valve **49** was closed; filtration was performed only by the hollow fiber membrane module **36**. Obtained recovery liquid was led out of the container **1** from the filtrate outlet **3**, sent to the filtrate tank **12** via the filtrate bypass pipe **24**, and recovered there. Pressure filtration was performed until there was no recovery liquid led out from the filtrate outlet **3** of the hollow fiber membrane module **36**, and the recovery step was finished.

(82) In this operation, the amount of liquid remaining finally inside the pipes etc. of the membrane filtration unit was 31 L. The collection step and the recovery step took 9 minutes.

Example 2

(83) The same membrane filtration unit operating method as in Example 1 was performed except that in the collection step the filtrate division valve **44** was closed and reverse flowing was caused in the hollow fiber membrane modules **31-35** when filtrate was caused to flow in the reverse direction from the secondary sides of the hollow fiber membranes to their primary sides.

(84) In this operation, the amount of liquid remaining finally inside the pipes etc. of the membrane

filtration unit was 31 L. The collection step and the recovery step took 7 minutes.

Example 3

(85) The same membrane filtration unit operating method as in Example 1 was performed except that the collection step and the recovery step were executed simultaneously. To execute the collection step and the recovery step simultaneously, reverse flowing was caused by introducing nitrogen gas of 150 kPa from the second pressurized gas introduction pipe **26** and, at the same time, pressure filtration was performed by introducing nitrogen gas of 100 kPa from the first pressurized gas introduction pipe **25**.

(86) In this operation, the amount of liquid remaining finally inside the pipes etc. of the membrane filtration unit was 31 L. The collection step and the recovery step took 6 minutes.

Example 4

(87) The same membrane filtration unit operating method as in Example 1 was performed except that a membrane filtration unit exemplified in FIG. 3 using six hollow fiber membrane modules prepared was constructed. The raw liquid inlet **2** of only the hollow fiber membrane module **36** was disposed below the lower end of the supply liquid pipe **21** in the vertical direction by 300 mm.

(88) In this operation, the amount of liquid remaining finally inside the pipes etc. of the membrane filtration unit was 9 L. The collection step and the recovery step took 14 minutes.

Comparative Example 1

(89) A membrane filtration unit exemplified in FIG. 2 was constructed by using six hollow fiber membrane modules prepared in such a manner that their longitudinal direction coincides with the vertical direction.

(90) The filtration step was executed by a crossflow filtration operation by sending raw liquid from the raw liquid tank **11** to the hollow fiber membrane modules **31-36** by the supply pump **13**. At this time, the supply liquid valve **41** was open, the filtrate valve **42** was open, the concentrate valve **43** was open, the filtrate division valve **44** was open, the filtrate bypass valve **45** was closed, the backwash valve **46** was closed, the backwash bypass valve **47** was open, the concentrate gas introduction valve **48** was closed, and the filtrate gas introduction valve **49** was closed. The crossflow flow rate and the filtration flow rate were set at 103 m.sup.3/hr and 7.7 m.sup.3/hr, respectively. Backwashing was performed in the middle of the filtration step to clean clogging of the hollow fiber membranes **5**. More specifically, filtrate liquid was sent to the hollow fiber membrane modules **31-36** from the filtrate tank **12** by the backwash pump **14**. At this time, the supply liquid valve **41** was closed, the filtrate valve **42** was closed, the concentrate valve **43** was open, the filtrate division valve **44** was open, the filtrate bypass valve **45** was closed, the backwash valve **46** was open, the backwash bypass valve **47** was open, the concentrate gas introduction valve **48** was closed, and the filtrate gas introduction valve **49** was closed. The backwash flow rate was set at 10.4 m.sup.3/hr.

(91) In the filtration step, a cycle consisting of the above-mentioned crossflow filtration operation for 9 minutes and the above-mentioned backwashing for 1 minute was performed repeatedly. The crossflow filtration operation was stopped and the filtration step was finished when the amount of raw liquid became small and liquid sending was completed to the lower limit level of the raw liquid tank **11**.

(92) Subsequently, the recovery step was executed without executing the collection step. In the recovery step, pressure filtration was performed by introducing nitrogen gas of 100 kPa from the first pressurized gas introduction pipe **25**. At this time, the supply liquid valve **41** was closed, the filtrate valve **42** was closed, the concentrate valve **43** was closed, the filtrate division valve **44** was closed, the filtrate bypass valve **45** was open, the backwash valve **46** was closed, the backwash bypass valve **47** was closed, the concentrate gas introduction valve **48** was open, and the filtrate gas introduction valve **49** was closed; filtration was performed only by the hollow fiber membrane module **36**. Obtained recovery liquid was sent to the filtrate tank **12** via the filtrate bypass pipe **24** and recovered there. Pressure filtration was performed until there was no recovery liquid led out

from the filtrate outlet 3 of the hollow fiber membrane module 36, and the recovery step was finished.

(93) In this operation, the amount of liquid remaining finally inside the pipes etc. of the membrane filtration unit was 55 L. The recovery step took 24 minutes.

(94) The present invention has been described in detail by referring to the particular embodiment, it is apparent to those skilled in the art that various changes and modifications are possible without departing from the spirit and scope of the invention.

(95) The operating method of a membrane filtration unit according to the present invention is applied preferably to membrane filtration of raw liquid in a water treatment field such as drinking water manufacture, water purification treatment, and effluent treatment, a fermentation field that involves cultivation of microbes or cultivation cells, and a food industry field.

DESCRIPTION OF SYMBOLS

(96) 1: Container 2: Raw liquid inlet 3: Filtrate outlet 4: Raw liquid outlet 5: Hollow fiber membranes 6: Raw liquid-side space (primary side) 7: Filtrate-side space (secondary side) 8: Potting portion 11: Raw liquid tank 12: Filtrate tank 13: Supply pump 14: Backwash pump 21: Supply liquid pipe 22: Filtrate pipe 23: Concentrate pipe 24: Filtrate bypass pipe 25: First pressurized gas introduction pipe 26: Second pressurized gas introduction pipe 27: Backwash pipe 28: Filtrate recovery pipe 31-36: Hollow fiber membrane modules 41: Supply liquid valve 42: Filtrate valve 43: Concentrate valve 44: Filtrate division valve 45: Filtrate bypass valve 46: Backwash valve 47: Backwash bypass valve 48: Concentrate gas introduction valve 49: Filtrate gas introduction valve

Claims

1. A method of operating a membrane filtration unit, the membrane filtration unit comprising a plurality of hollow fiber membrane modules connected to each other in parallel, each hollow fiber membrane module comprising a container which has a raw liquid inlet, a filtrate outlet, and a raw liquid outlet and is filled with hollow fiber membranes, the method comprising: a filtration step of introducing a raw liquid into the container from the raw liquid inlet, filtering the raw liquid through the hollow fiber membranes from a primary side to a secondary side, and leading out resulting filtrate to outside the container from the filtrate outlet; a collection step of causing the filtrate existing in the container to flow in a reverse direction through the hollow fiber membranes from the secondary side to the primary side and leading out resulting backflow liquid to outside the container from at least one of the raw liquid inlet and the raw liquid outlet; and a recovery step of introducing the backflow liquid into the container again from at least one of the raw liquid inlet and the raw liquid outlet, filtering the backflow liquid through the hollow fiber membranes from the primary side to the secondary side, and leading out resulting recovery liquid to outside the container from the filtrate outlet, wherein a relation of $n_{sub.1} \geq n_{sub.2} > n_{sub.3}$ is satisfied, where $n_{sub.1}$ is the number of the hollow fiber membrane modules simultaneously used in executing the filtration step, $n_{sub.2}$ is the number of the hollow fiber membrane modules simultaneously used in executing the collection step, and $n_{sub.3}$ is the number of the hollow fiber membrane modules simultaneously used in executing the recovery step; wherein the filtration step introduces raw liquid through $n_{sub.1}$ hollow fiber membrane modules, the collection step introduces filtrate into $n_{sub.2}$ hollow fiber membrane modules, and the recovery step introduces backflow liquid into $n_{sub.3}$ hollow fiber membrane modules.

2. The method of operating a membrane filtration unit according to claim 1, wherein at least one of the hollow fiber membrane modules is used in executing the recovery step and at least one of the hollow fiber membrane modules is used in executing the collection step, wherein the at least one hollow fiber membrane module used in executing the recovery step is different from the at least one hollow fiber membrane module used in executing the collection step.

3. The method of operating a membrane filtration unit according to claim 1, wherein the collection step and the recovery step are executed using pressurized gas and a pressure P2 of the pressurized gas used in executing the collection step and a pressure P3 of the pressurized gas used in executing the recovery step satisfy a relation of $P3 < P2$.
 4. The method of operating a membrane filtration unit according to claim 2, wherein the raw liquid inlet of the at least one of the hollow fiber membrane modules used in executing the recovery step is located below the raw liquid inlet of at least one of the hollow fiber membrane modules that is not used in executing the recovery step in a vertical direction.
 5. The method of operating a membrane filtration unit according to claim 3, wherein the hollow fiber membranes are membranes made of a hydrophobic resin.
 6. The method of operating a membrane filtration unit according to claim 5, wherein the hydrophobic resin comprises polyvinylidene difluoride.
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